Part One

ROOTS

This book begins in Miletus, twenty-six centuries ago. Why begin a book about quantum gravity with events, people and ideas so ancient? I hope the reader eager to get on to quanta of space will not hold this against me. For it is easier to understand ideas by starting with the roots from which they have grown, and an important number of the ideas which turned out to be effective for understanding the world originated over two thousand years ago. If we briefly retrace their birth, they become clearer, and the later steps turn out to be simpler and natural.

But there's more. Certain problems first posed in antiquity continue to be crucial to our understanding of the world. Some of the most recent ideas about the structure of space utilize concepts and issues introduced then. In speaking of these distant ideas, I put on to the table questions which are going to be central to quantum gravity. This makes it also possible, when treating of quantum gravity, to distinguish between the ideas which go back to the very origin of scientific thought, even if we are unfamiliar with them, and those which are radically new. The connection between problems posed by the scientists of antiquity, and solutions found by Einstein and quantum gravity, is, as we shall see, surprisingly close.